

Ravindranath Tagore

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CAREER OBJECTIVE

Seeking an opportunity to apply my problem-solving and analytical skills to deliver practical solutions and contribute to impactful business outcomes.

EDUCATION

Currently pursuing a Bachelor of Engineering in **Information Science and Engineering** (4th year) at Dayananda Sagar Academy of Technology and Management (DSATM) Bengaluru, Karnataka with a **CGPA of 8.23**

SKILLS

Programming Languages: Java, Python, JavaScript, SQL

Web Development: HTML, CSS, React

Design Tools: Canva

Tools & Platforms: Git, GitHub, Visual Studio Code, IntelliJ IDEA

Soft Skills: Leadership, Creativity, Problem-Solving

PROJECTS

1. Slates.me - Canvas Collaboration Platform | [Live Site](#)

- Built **Slates.me** using **React** for the frontend and Supabase for backend data storage and **real-time collaboration**
- Implemented a **shared canvas feature** where users can add and move notes and files (**images, videos, PDFs**)
- Made a **responsive UI** with gesture support, auto-save, and easy access from any device

2. PhysioSense - Gamified Wrist Mobility Rehabilitation System | [GitHub](#)

- Engineered a **gamified wrist mobility rehabilitation system** utilizing **ESP8266 microcontroller** and **MPU6050 Sensor**
- Implemented a **Python Flask server** using Socket communication to process **wrist motion data** in real time and simulate keyboard inputs for **interactive game control**
- Enhanced **patient engagement** and **rehabilitation** outcomes through gamified wrist exercises

3. AI-Powered Orbital Object Detection and Classification System | [GitHub](#)

- Developed an advanced **space debris detection system** using **Python** and **YOLOv8** that analyzes satellite imagery in real-time through a web interface to identify and classify 11 types of space objects including satellites (CHEOPS, SOHO, SMART-1) and orbital debris with precise bounding box localization
- Implemented hybrid neural network architecture combining YOLO object detection with Vision Transformer components and trained the model on **space debris dataset**, achieving **86.5% mAP** accuracy, **85.1%** precision, and **76.7%** recall in detecting potential collision hazards for space safety applications

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) | [Certificate](#)

June 2025

Gained foundational knowledge of cloud concepts, core Azure services, security, and pricing.

ACHIEVEMENTS

- Received the **Local Award at NASA Space Apps Challenge 2024** (DSU Local Host) for demonstrating innovation and potential impact among **509+ teams**.
- 1st Prize in Inter-College Innovative Idea and Model Competition** - Dayananda Sagar Academy of Technology and Management, Bangalore.
- Second Runner-Up** in the **State-Level Project Exhibition 2025** - Bangalore Institute of Technology.
- Second Runner-Up** in the **Inter-College Frontend Development Competition** – Dayananda Sagar Academy of Technology and Management, Bangalore.
- Consolation Prize Winner** in **Altair Data Science Contest** (Feb–Apr 2025) - Organized by Department of Management Sciences (DoMS), IIT Kanpur, using Altair RapidMiner

EXTRACURRICULAR ACTIVITIES

- Core Member of Innovisions Technical Club, ISE Department
- Coordinated Technical Event TechnoSpark 2024
- Member of the National Service Scheme (NSS) since 2022